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The Risk of Cryptocurrency Payment Adoption and the Role of Social Media: Evidence From Online Travel Agencies

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Abstract

The swift advancement of social technologies has created unprecedented opportunities for companies to embrace new business models or bolster their existing ones. However, the adoption of controversial technologies with social characteristics can also expose firms to risks, as these technologies may not be entirely under their operational control. In this study, our objective is to analyze the impact of the adoption of a controversial technology (i.e., cryptocurrency payment) on users' perception of a firm and firm performance and examine how social media influences this impact. We leverage a unique research context where an online travel agency cooperates with Travala.com for cryptocurrency payment. Using conventional and synthetic difference-in-differences and randomized experiments, we compare the general perceptions of customers towards the online travel agency before and after the cooperation and gain three main findings. First, the adoption of cryptocurrency payment generally leads to a significant decrease in revenue because of the negative associations concerning cryptocurrency. Second, the popularity of cryptocurrency payment on social media mitigates this negative effect. Third, when cryptocurrency prices rise and public opinions on cryptocurrency turn positive, the mitigation effect is more salient. Further analysis reveals that our results are robust after controlling for the seasonality effect and the nature of the collaboration and using different lags for the outcome variable. This study is among the first to empirically examine how cryptocurrency payment adoption affects firm performance, and provide important theoretical and practical implications regarding the adoption of controversial technologies and the externalities of social media.

Keywords

Cryptocurrency payment, controversial technology, sales performance, social media, synthetic DID

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1 Introduction

Social technologies, for example, social media, live streaming, and crowdsourcing, have revolutionized business models and business operations (Cui et al., 2018; Huang et al., 2021; Khurana et al., 2019; Mallipeddi et al., 2022; Qiu et al., 2021) and reshaped consumer behavior (Qiu and Kumar, 2017; Qiu and Whinston, 2017). With the power of social media, individuals and businesses can easily discuss, promote, and educate others about cryptocurrencies and their benefits, creating a network effect that drives adoption and usage. Since 2020, a growing number of businesses have been embracing cryptocurrency payment, including PayPal, Visa, Microsoft, and Tesla (Shepherd, 2021); as of January 2024, more than 32,431 businesses, retailers, ATMs, and other infrastructures accepted cryptocurrency payment.[1]

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Despite the potential benefits of adopting cryptocurrency payment, including operational benefits and advertising effects, it may also have negative implications because of its controversial social reputation. There is uncertainty regarding the payoff of this technology, raising questions about its long-term viability and potential risks (Polasik et al., 2015). Many people believe that cryptocurrency has no intrinsic value (Smith, 2021) and is mainly used for illegal transactions and anonymous activities (Chu et al., 2019; Foley et al., 2019). Although supporters such as Elon Musk advocate cryptocurrency and cryptocurrency payment, Warren Buffett said that "Bitcoin is probably rat poison squared" (Kim, 2018) and Charlie Munger claimed that "Bitcoin is disgusting and contrary to the interests of civilization" (Li, 2021). In addition, its high volatility (Chu et al., 2019) and highly concentrated ownership and mining capacity (Makarov and Schoar, 2021) impede its acceptance. Therefore, the adoption of cryptocurrency payment may cause consumers to develop negative associations regarding the adopter.

Social media has transformed the way people communicate, share information, and interact with each other (Chen et al., 2014; Petrova et al., 2021). As social media platforms continue to evolve, they may play a role in changing users' perceptions of new and potentially controversial technologies (Etter et al., 2019). One such controversial technology is cryptocurrency payment, which has gained popularity in recent years with the rapid development of blockchain as a decentralized alternative to traditional payment methods (Milian et al., 2019).

Extant studies on cryptocurrency payment mainly investigated factors that influence consumer and company acceptance of cryptocurrency payment through surveys or experiments (Alshamsi and Andras, 2019; Borch, 2021; Catalini and Tucker, 2017; Jonker, 2019; Önder and Treiblmaier, 2018). To the best of our knowledge, no research has examined the effect of adopting cryptocurrency payment on firm performance nor how social media can influence people's general perceptions of such an adoption. As more and more companies begin to explore the adoption of cryptocurrency payment, it is crucial to understand the potential impact of this adoption on various dimensions of firm performance, such as revenue and market value. Furthermore, examining the role of social media in shaping people's perceptions of cryptocurrency payment adoption can provide valuable insights into how firms can effectively communicate and engage with stakeholders to mitigate potential risks and maximize benefits. By exploring the impact of cryptocurrency payment adoption and the role of social media in shaping perceptions, we can gain a better understanding of the opportunities and challenges associated with this emerging technology and develop strategies to effectively navigate the rapidly changing landscape of digital payment systems, and more importantly, how to use social media to mold the way people perceive the implementation of controversial technologies. Therefore, in this research, we focus on the following two questions: (i) How does cryptocurrency payment adoption affect firm performance? (ii) How does social media moderate the effect of adopting this controversial technology?

We answer the above research questions by analyzing the adoption of cryptocurrency payment among online travel agencies (OTAs). The adoption took place in collaboration with a cryptocurrency payment platform that specializes in the travel industry (Travala.com). We collect consumer reviews as a proxy for OTA firm performance from all 51 states of the United States and employ sets of difference-in-differences (DID) designs (i.e., conventional DID and synthetic DID) to empirically examine whether the performance of the treatment group increases or decreases after the adoption of cryptocurrency payment relative to the control group. We further explore the moderating impact of social media on this effect to understand the potential underlying mechanisms. In order to compensate for the limitations of relying solely on observational data and for endogeneity issues, such as the potential consumer flow from the treatment to the control groups, and to further strengthen the reliability and validity of our findings, we also conduct randomized experiments.

We find that the adoption of cryptocurrency payment is negatively associated with the firm performance of the adopter, and this negative relationship is mitigated by the popularity of the technology provider (i.e., Travala.com) on social media. These results are robust when different time lags are used to calculate the outcome variable and are substantiated via randomized experiments. We also rule out potential alternative explanations such as the heterogeneous seasonal effect and the effect of cooperating with small-brand companies. Finally, we demonstrate the moderating effect of cryptocurrency price to establish the underlying mechanism of the negative associations.

This study contributes to the literature on social technology adoption in various ways. Existing research mainly demonstrates the positive effect of social technologies such as social promotion and social media customer services (Gao et al., 2020; Huang et al., 2021). Our study adds to this stream of research by revealing the negative effect of adopting a controversial technology—cryptocurrency payment—and investigating the mitigating impact of social media. Moreover, we extend the theoretical application of the associative learning model into the field of new technology adoption. We also contribute to the literature on cryptocurrency payment adoption by demonstrating the direct effect of adoption on firm performance based on a secondary dataset with rigorous causal inference methods and additional randomized experiments, rather than focusing only on factors influencing the acceptance of this technology, as in previous studies (Jonker, 2019; Polasik et al., 2015; Roussou et al., 2019). From a practical perspective, our study underscores the importance of carefully considering the potential impact of cryptocurrency payment adoption on customer perceptions and developing strategies to address any negative effects.

2 Literature Review and Theoretical Foundation

2.1 Perceptions of Cryptocurrency Payment

2.1.1 Recent Studies. Extant research on the adoption of cryptocurrency payment mainly focused on factors that may affect consumers and firm acceptance. Few studies have considered the impact of cryptocurrency payment adoption on firm performance. For example, based on the technology acceptance model, Mendoza-Tello et al. (2019) and Roussou et al. (2019) investigated the factors that may influence user acceptance of cryptocurrency payment and found that perceived usefulness, security, and compatibility are determining factors. Jonker (2019) collected survey data about cryptocurrency payment adoption from a large representative sample of retailers in the Netherlands and found that these retailers are interested in adopting cryptocurrency, but the lack of consumer demand, transactional benefits, and perceived accessibility are barriers to cryptocurrency acceptance. Other studies have determined that early adopters play an important role in the diffusion of cryptocurrency (Catalini and Tucker, 2017) and speculative price bubbles may accelerate the diffusion and adoption of cryptocurrency (Wei and Dukes, 2021).

2.1.2 Social Characteristics. As a novel payment technology, cryptocurrency payment has several unique social characteristics. First, its infrastructure is based on decentralized peer-to-peer distributed networks (Babich and Hilary, 2020). Since the technology relies on decentralized communities of users, social media could have a significant impact on the value and acceptance of cryptocurrency and cryptocurrency payments. For example, Elon Musk has a tremendous influence on the cryptocurrency market through his Twitter account (Kartsev, 2021), and his posts are known to cause a frenzy on Dogecoin (Kharpal, 2021). Mai et al. (2018) found that the effect of social media on the price of Bitcoin is mainly driven by the silent majority, and information on forums has a stronger impact on future Bitcoin value than information from tweets. Xie et al. (2020) determined that less cohesive networks on social forums are better at predicting future Bitcoin returns than more cohesive networks. As many blockchain-related communities have transformed into decentralized autonomous communities (Beck et al., 2018; Hassan and De Filippi, 2021; Wang et al., 2019), decisions about technology development are not controlled by adopting firms but by development communities.

2.1.3 Associative Learning. The adoption of controversial technologies with social characteristics not only affects the adopter but can also have far-reaching impacts on consumers (Besharat and Langan, 2014; Kalaignanam et al., 2007). These effects are often attributed to consumers' associations (Besharat and Langan, 2014), which develop through associative learning (Shimp, 1991; Washburn et al., 2004). Associative learning refers to instances when consumers make connections between events that take place in their environment (Shimp, 1991). There are two types of associative learning: *instrumental learning* and *classical conditioning*. Instrumental learning refers to consumers developing positive or negative attitudes toward a firm that has adopted a new technology based on their experiences with the technology. Classical conditioning refers to consumers pairing new technology adoption with cues that elicit a positive or negative response without actually experiencing the technology. The associative learning model has been widely applied in brand alliance studies (Van Osselaer and Janiszewski, 2001). Van Osselaer and Alba (2000) suggested that consumers predict high product quality if they have a high-quality perception of both the brand name and the predictive product attribute. Van Osselaer and Janiszewski (2001) used the associative learning framework to explain the effect of brand names on product extensions and brand alliances.

In our research context, while some firms are starting to accept cryptocurrency payment (Shepherd, 2021) and many consumers are familiar with cryptocurrency (Ballard, 2019), the overall population that uses cryptocurrency payment is small. Therefore, according to the associative learning model, the effect of cryptocurrency payment lies in classical conditioning. That is, the cues paired with cryptocurrency elicit users' associations with cryptocurrency payment, which affect users' attitudes toward firms that adopt it. Hence, there are two possible consequences of firms adopting this controversial technology. (i) The adoption could cause a positive association because it opens up a new channel for payment with attractive features such as safety and speed. This positive association will make consumers develop a favorable attitude toward the adopter and improve its sales performance. (ii) The adoption could cause a negative association because it involves security and operational risks. This negative association will make consumers develop a negative attitude toward the adopter and undermine its sales performance. Given the potential negative outcomes of cryptocurrency payment adoption, it is important to identify the solutions that can be used to mitigate customers' negative associations.

2.2 Social Media Externalities

Traditional payment methods such as cash and credit cards are mature technologies that firms are well equipped to handle. In contrast to these traditional payment methods, the adoption of cryptocurrency payment often attracts social media attention due to its social characteristics. As a controversial technology, cryptocurrency payment possesses significant social characteristics that can affect consumer attitudes to the technology when more people use it. These attitudes may be shared on social media and influence others' perceptions, which can further impact the adopter's performance. Therefore, we propose that social media can moderate individuals' associations with this technology, potentially changing customers' negative associations and serving as a solution to address negative perceptions.

While social media has brought many benefits, such as increased connectivity, access to information, and new opportunities for expression and creativity, it has also given rise to a range of externalities that can have both positive and negative effects on individuals and society (Kumar et al., 2018; Lee et al., 2018). One of the most notable externalities of social media is the way it can amplify and spread information (Clarke et al., 2020; Kranova et al., 2015; Mayzlin et al., 2014). The moderating effect of social media has been widely reported in prior studies. The consensus is that social media is a double-edged sword. Positive social media attention can expedite a firm's success, while negative social media attention can accelerate a firm's downfall. The GameStop trading mania is a good example of the power of social media in amplifying the effect of online investment communities on stock prices (MacMillan and Torbati, 2021). Similarly, in the context of online review platforms, Ye et al. (2009) found that consumer reviews have a significant effect on hotel sales, but the effect can be moderated by managerial responses (Gu and Ye, 2014). In the context of political impact, social media can make or break a politician's career and influence voting patterns (Mousavi and Gu, 2019; Petrova et al., 2021).

Existing research documents both positive and negative externalities of social media; however, the role of social media in mitigating or exacerbating the impacts of adopting controversial technologies is unclear, which motivates us to conduct this research. In the context of this study, we expect that the popularity of Travala.com on social media and the recognition of Travala.com or cryptocurrency payment among social media users play a positive role in mitigating consumers' negative associations with controversial technologies. That is, for consumers with negative attitudes or perceptions towards cryptocurrency payment, the promotion and spread of cryptocurrency payment on social media has the potential to relieve consumers' concerns about the risk of cryptocurrency payment. For consumers with positive or neutral attitudes towards cryptocurrency payment, social media can amplify the benefit of cryptocurrency payment.

3 Empirical Settings

3.1 Research Context

Travala.com is a cryptocurrency platform that specializes in the travel industry. It announced its first cooperation with Booking.com on 25 November 2019. Since then, Travala.com has established partnerships with other popular OTAs, such as Priceline, Expedia, and Agoda. Table 1 lists five cooperation events reported by Travala.com. In this study, we choose to analyze the first event for two reasons. First, the research context of the first cooperation between Travala.com and Booking.com is clean, but there is extensive overlap or repeated treatment across the other cooperations. Second, events 2, 3, and 4 were seriously affected by COVID-19.

The start of the cooperation between Travala.com and Booking.com was widely reported by the mainstream media, including Yahoo! Finance, and no other important events involving Booking.com were observed on other media (e.g., Seeking Alpha) during the event window.[2] The cooperation allows all the hotel listings on Booking.com to be available via Travala.com. That is, consumers can book all the hotels listed on Booking.com via Travala.com using cryptocurrency payment, but consumers cannot directly use cryptocurrency to book hotels on Booking.com. Moreover, consumers cannot post reviews on Booking.com for reservations that they made through Travala.com. This unique context enables us to observe the change in online sales proxied by the number of consumer reviews on Booking.com before and after its cooperation with Travala.com. In order to perform a rigorous empirical study, we utilize DID designs (i.e., conventional DID and synthetic DID) that use the same set of hotels listed on Expedia as the control group (unaffected by the event[3]). As Expedia and Booking.com only allow consumers who have made reservations to post reviews on their websites, the DID designs capture the treatment effect of the cooperation with Travala.com on the online sales of Booking.com.

Although the decrease in sales on Booking.com could be compensated by the increase in sales on Travala.com via cryptocurrency payment, the sales of Travala.com are negligible compared to the sales of Booking.com (see Table 2). With regard to sales, it should be noted that the number of room nights of Booking Holdings includes Booking.com (the treatment group), Priceline, and Agoda, whereas the number of room nights of Expedia Group (the control group) includes Expedia, Hotels.com, Orbitz, and Travelocity. The statistics in Table 2 were obtained from Travala.com and the financial reports of Booking Holdings and Expedia Group.

3.2 Data and Summary Statistics

First, we collected all the consumer reviews of hotels listed on Booking.com (excluding the reviews on Priceline and Agoda) and Expedia Group (including reviews on Expedia, Hotels.com, Orbitz, and Travelocity) in all 51 states of the United States. We then matched these hotels between Booking.com and Expedia Group using their names and addresses and identified the same set of hotels listed on Expedia Group as the control group. Table 3 reports the summary statistics and sample distribution of the dataset. The number of reviews for each hotel in each week is the dependent variable in this study. The average number of reviews per week (16 weeks before and after the event) for the treatment group (Booking.com) is about 2.54, while that for the control group (Expedia Group) is 2.58. Among the Expedia Group platforms, Expedia and Hotels.com account for the majority of consumer reviews.

We also retrieved the number of tweets posted weekly by Travala.com (*Travala-post*), the number of tweets mentioning "Travala" (*Travala-include*), as well as the number of likes (*Travala-post-like* and *Travala-include-like*) these

Table 1. The timeline of cooperation events.

No.	Event	Time	News
1	Travala.com & Booking.com Strategic Partnership	2019/11/25	https://blog.travala.com/travala-com-booking-com-strategic-partnership-targets-massive-crypto-adoption/
2	Travala.com Partners With Leading Travel Provider, Priceline	2020/03/07	https://blog.travala.com/travala-com-partners-with-leading-travel-provider-priceline/
3	Expedia and Travala.com Join Forces to Offer Frictionless Cryptocurrency Travel Booking	2020/07/06	https://blog.travala.com/expedia-and-travala-com-join-forces-to-offer-frictionless-cryptocurrency-travel-booking/
4	Travala.com and Agoda Partner to Boost Travel with Bitcoin and other Cryptocurrencies	2020/08/03	https://blog.travala.com/travala-com-and-agoda-partner-to-boost-travel-with-bitcoin-and-other-cryptocurrencies/
5	Travala.com Signs Enhanced Partnership with Expedia Partner Solutions	2021/03/19	https://blog.travala.com/travala-com-signs-enhanced-partnership-with-expedia-eps/

Table 2. Sales made on Travala.com versus Booking.com three months before and after November 2019.

Month	Travala.com (room nights)	Quarter	Booking holdings (room nights)	Expedia group (room nights)
2019/08	1,045	2019/Q3	223 millions	116.5 millions
2019/09	1,009			
2019/10	1,105	2019/Q4	191 millions	91.6 millions
2019/11	1,326			
2019/12	1,528			
2020/01	1,755	2020/Q1	124 millions	69.4 millions
2020/02	2,364			

Table 3. Summary statistics.

Category	Variable	Mean	SD	Min.	Max.
Weekly RevNum	Booking.com	2.5361	5.6210	0	350
	Expedia group	2.5822	5.1102	0	362
	Expedia	1.2775	3.0152	0	213
	Hotels.com	1.1024	2.0440	0	107
	Orbitz	0.1202	0.4449	0	27
	Travelocity	0.0767	0.3189	0	17
Social Media	*Travala-post*	17.5000	8.0763	4	39
	Travala-include	191.2188	120.6107	66	474
	Travala-post-like	641.3125	373.9186	137	1842
	Travala-include-like	1243.3750	733.9372	392	3388
Crypto Price	*Mega*	−0.0121	0.0905	−0.3169	0.2214
	Large	−0.0129	0.0913	−0.3166	0.2163
	Broad	−0.0128	0.0920	−0.3183	0.2175
	AVA	0.0078	0.1811	−0.6064	0.4079

tweets received. These variables reflect the social influence, popularity, and netizen recognition of Travala.com (a leading cryptocurrency payment OTA) and function as the moderating variables. We observe that the weekly number of tweets mentioning “Travala” is more than ten times greater than the number of tweets posted by Travala.com, while the weekly number of likes of tweets mentioning “Travala” is about two times that of tweets posted by Travala.com.

Another set of moderating variables is cryptocurrency price trends (*CryptoPrice*), which is measured by $\ln(\text{price}_t/\text{price}_{t-1})$. We use up to four indexes to measure the trend of cryptocurrency prices. First, the cryptocurrency mega-cap index (*Mega*) follows the performance of Bitcoin and Ethereum. Second, the cryptocurrency large-cap index (*Large*) follows the performance of cryptocurrencies with the largest market capitalizations. Third, the cryptocurrency broad digital market index (*Broad*) follows the performance of digital assets satisfying certain liquidity and market capitalization requirements detailed in its eligibility criteria. Fourth, the price changes of *AVA*, which is offered by Travala.com. We observe that in the event window (four months before and after the event), the average weekly returns of all the indexes (apart from *AVA*) are negative. The price volatility of *AVA* is also higher than that of the other three indexes.

3.3 Research Design

3.3.1 Conventional DID Identification. We use the conventional DID design to estimate the effect of cooperation with Travala.com on the hotel sales of Booking.com. In Model (1), *i* denotes hotel, *t* denotes week, and *p* denotes OTA. The dependent variable is the log value of the weekly number of reviews posted about hotel *i* on OTA *p* in time period *t* (*LogRevNum*). *Treat* is a dummy variable to denote whether the platform is treated, which is set to 1 for reviews posted on Booking.com and 0 for those posted on Expedia Group platforms. *Post* is a dummy variable to denote whether the reviews were posted after (*Post* = 1) or before (*Post* = 0) on 25 November 2019. We also include the hotel-fixed effect and the time-fixed effect in Model (1) to control for hotel heterogeneity and time-variant factors. We further control for website-level and state-level trends by including the lagged cumulative number of hotel reviews for each state per website. The coefficient of the interaction term *Treat*Post* (for brevity, we call *Treat*Post Treated* hereafter) captures the treatment effect of cooperation with Travala.com on the hotel sales of Booking.com. In order to obtain robust empirical results, we use up to four time windows (4, 8, 12, and 16 weeks before and after 25 November 2019).

$$LogRevNum_{it|p} = \beta_0 + \beta_1 Treat_{i|p} \times Post_t + U_{i|p} + V_t + Control + \xi_{it|p}. \tag{1}$$

3.3.2 Synthetic DID Identification. Conventional DID estimation has been used in many studies (Moser and Voena, 2012). It is usually applied when there is a substantial number of treated units and when the underlying trends of the treatment group and the control group are assumed to be parallel (Angrist and Pischke, 2009). Another widely applied method, the synthetic control method (SC), weights the outcomes of the control group to predict the outcomes of the treatment group assuming that it is not treated (Abadie and Gardeazabal, 2003). The method optimally calculates the weight parameter to match the pre-event trend of the treatment group; therefore, the parallel trend assumption is not required. The treatment effect can thus be estimated using the difference between the observed post-event outcomes and the predicted post-event outcomes (Abadie, 2021; Abadie et al., 2010). However, SC is usually used in cases that have only a single treated unit or a small number of treated units and requires long and balanced pre-adoption panels.

Given that our research has a large number of treatment units and the potential violation of parallel trend assumptions might lead to a biased estimation, we use a newly proposed method, synthetic DID (SDID hereafter) (Arkhangelsky et al., 2021), to add to the rigorousness of our study. Arkhangelsky et al. (2021) combined the attractive features of both DID and SC, stating that “like SC, our method re-weights and matches pre-exposure trends to weaken the reliance on parallel trend type assumptions. Like DID, our method is invariant to additive unit-level shifts and allows for valid large-panel inference.”

$$(\hat{\tau}_{did}, \hat{u}, \hat{\alpha}, \hat{\beta}) = \underset{\alpha,\beta,\mu,\tau}{\arg\min} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - u - \alpha_i - \beta_t - W_{it}\tau)^2 \right\}, \tag{2}$$

$$(\hat{\tau}_{sc}, \hat{u}, \hat{\beta}) = \underset{\mu,\beta,\tau}{\arg\min} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - u - \beta_t - W_{it}\tau)^2 \, \hat{\omega}_i^{sc} \right\}, \tag{3}$$

$$(\hat{\tau}_{sdid}, \hat{u}, \hat{\alpha}, \hat{\beta}) = \underset{\alpha,\beta,\mu,\tau}{\arg\min} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - u - \alpha_i - \beta_t - W_{it}\tau)^2 \, \hat{\omega}_i^{sdid} \, \hat{\lambda}_t^{sdid} \right\}. \tag{4}$$

Equations (2) to (4) illustrate the average treatment effect (denoted by τ) of different estimation methods (DID, SC, and SDID, respectively), in which μ is the model concept, α is the unit fixed effect, and β is the time fixed effect. *N*, *T*, *Y*, and *W* denote the unit number, time periods, the outcome of a unit, and whether the unit is exposed to treatment, respectively. ω is the unit weight and γ is the time weight. The unit weights align pre-exposure trends in the outcomes of treated and control units, while the time weights balance pre-exposure time periods with post-exposure ones. Pre-exposure periods that are more predictive of post-exposure outcomes receive higher weights. According to these equations, the DID estimator includes the two-way fixed effect (unit and time fixed effects) without either time or unit weights. Although the SC estimator includes unit weights, it omits the unit fixed effect and the time weights. The SDID estimator includes the two-way fixed effect (unit and time fixed effects) and considers the time and unit weights. That is, SDID puts more weight on units that are similar in terms of their past to the treated units and emphasizes periods that are similar to the target periods.

We thus utilize the SDID design to check the robustness of the DID estimation results by relaxing the requirements of the parallel trend assumption (Berman and Israeli, 2022). Within the SDID design, the outcome variable is the log value of the weekly number of consumer reviews (*LogRevNum*), the unit indicator is the hotel ID, and the time indicator is the week order. Regarding the treatment indicator (a dummy variable to denote the treatment group), we set the value to 1 for observations of Booking.com after the event and set the value to 0 for other observations (i.e., observations of Booking.com before the event and observations of multiple OTAs in Expedia Group). Arkhangelsky et al. (2021) indicated that the conditions (including the correlation between treatment assignments, unit-level time trends, and heterogeneous treatment effects) of getting a consistent and asymptotically normal estimator of τ are quite general. The important condition is that the combined number of control units and pre-adoption

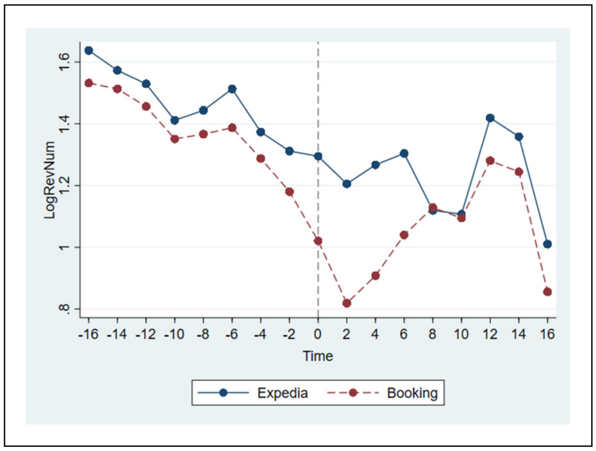

Figure 1. Trends of *LogRevNum*.

periods is sufficiently large compared to the combined number of treated units and post-adoption periods. The condition is well satisfied in our research context, since we separated consumer reviews on Expedia Group platforms (i.e., Expedia, Hotels.com, Orbitz, and Travelocity) in the control group to generate more control units.

4 Results

4.1 DID and SDID Estimation Results

Before performing the DID estimation, we first depict the trends of *LogRevNum* for both Booking.com (red) and Expedia Group (blue) in Figure 1. We observe that the pre-event trends roughly satisfy the parallel trend assumption. In order to alleviate the concern that imperfect parallel trends may threaten our estimation, we first perform conventional DID estimation and then use SDID estimation to illustrate the validity of our results. Table 4 presents the DID estimation results of Model (1) using four time windows (i.e., 4, 8, 12, and 16 weeks before and after event 1). We observe that the coefficients of *Treated* are significantly negative across all time windows. In column 1 (4 weeks before and after event 1), the magnitude of the coefficient is −0.169, indicating that the cooperation with Travala.com leads to about a 15.5% (1−exp(−0.169)) decrease in online sales on Booking.com. However, this negative consequence weakens over time, as shown in columns two to four. These results indicate that the cooperation with Travala.com exerts a short-term negative effect on the platform sales of Booking.com.

Table 5 presents the SDID estimation results. Although the magnitude of the coefficients is lower than that of those in Table 4 (DID), their sign and significance remain unchanged. These results further confirm the robustness of the negative treatment effect induced by the event. In column 1 of Table 5, the magnitude of the treatment effect is −0.139, indicating that the cooperation with Travala.com leads to a 13% (1−exp(−0.139)) decrease in online sales on Booking.com. If we focus on the time window of four months (i.e., 16 weeks) before and after the event, the treatment effect is about 3.5% (1−exp(−0.0354)). These results indicate that the cooperation with Travala.com exerts a short-term negative effect on the platform sales of Booking.com. As Figure 2a to d illustrates, the pre-event parallel trends are very good after re-weighting by the SDID estimation.

4.2 The Externalities of Social Media

In order to examine how social media drives the adoption of cryptocurrency payment and its influence on the adoption effect, we retrieved the weekly number of tweets posted by Travala.com (*Travala-post*), tweets mentioning “Travala” (*Travala-include*), and likes these tweets received. Tweets posted by Travala.com and their likes denote the social influence of Travala.com on Twitter and the recognition from Twitter users, respectively. Tweets posted by Twitter users mentioning “Travala” and their likes indicate the popularity of Travala.com on Twitter, a leading social media platform. We then interact these variables with *Treated* as in Model (1) to examine the moderating effect of social media. Table 6 reports the estimation results. We observe that although the coefficient of *Treated* is significantly negative, all the coefficients of the interaction terms are positive and significant. These results suggest that the social influence and popularity of Travala.com positively moderate the negative effect of cryptocurrency payment adoption.

4.3 The Moderating Effect of Cryptocurrency Price

We use four measures (*Mega*, *Large*, *Broad*, and *AVA*) to reflect cryptocurrency price trends and explore whether the decrease in room sales on Booking.com induced by the partnership with Travala.com is mitigated when cryptocurrency prices rise. We conjecture that consumers’ sentiments regarding cryptocurrency are influenced by cryptocurrency price changes. We thus interact these variables with *Treated* as in Model (1) to examine the moderating effect of cryptocurrency prices. As observed in Table 7, no matter which cryptocurrency price index is utilized, the coefficients of the interaction term are all positive and significant. These results indicate that the sales decrease on Booking.com after its cooperation with Travala.com is more severe when the cryptocurrency price is lower, and this negative effect is mitigated when the cryptocurrency price rises.

5 Robustness Check

5.1 Using Different Lags to Calculate the Dependent Variable

Although we have obtained information about the review date and length of stay related to each consumer review, the check-in date related to each review remains inconclusive, because consumers may not post reviews immediately after their stays

Table 4. Difference-in-differences (DID) estimation results.

Column	(1)	(2)	(3)	(4)
Time window	[−4, 4] weeks	[−8, 8] weeks	[−12, 12] weeks	[−16, 16] weeks
Treated	−0.1685***	−0.0646***	−0.0260***	−0.0523***
	(0.0034)	(0.0030)	(0.0029)	(0.0029)
State-website trends	0.0044***	0.0050***	0.0084***	0.0079***
	(0.0004)	(0.0003)	(0.0002)	(0.0002)
Hotel fixed effect	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes
Obs.	487,408	974,816	1,462,224	1,949,632
R^2	0.7017	0.6697	0.6524	0.6397

Note: Robust standard errors (clustered by hotel ID) in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 5. Synthetic difference-in-differences (SDID) estimation results.

Column	(1)	(2)	(3)	(4)
Time window	[−4, 4] weeks	[−8, 8] weeks	[−12, 12] weeks	[−16, 16] weeks
Coef.	−0.1393***	−0.0814***	−0.0466***	−0.0354***
SE	(0.0044)	(0.0037)	(0.0035)	(0.0034)
t	−31.5704	−22.0068	−13.4737	−10.4568
Hotel fixed effect	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

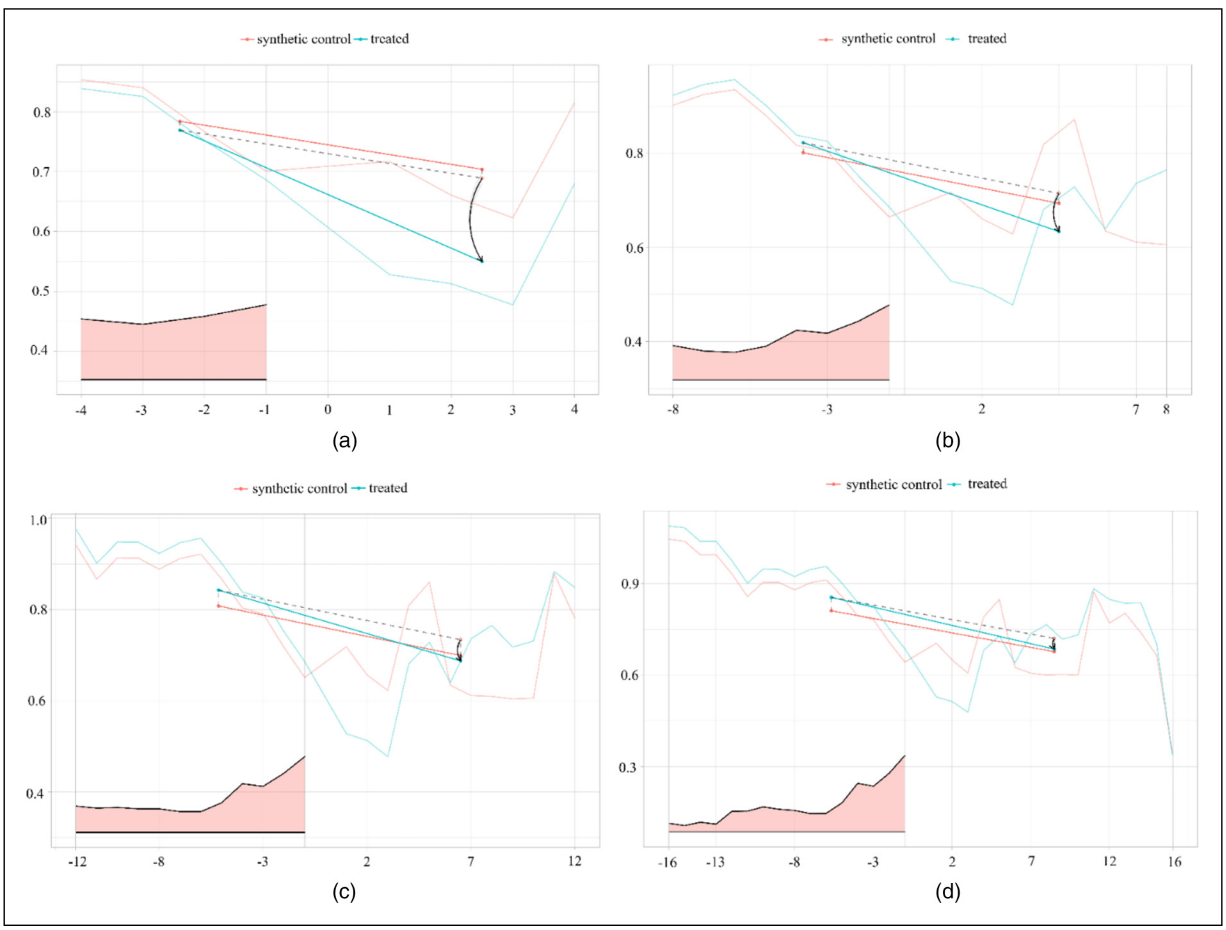

Figure 2. (a) Treatment effect ([−4, 4] weeks), (b) treatment effect ([−8, 8] weeks), (c) treatment effect ([−12, 12] weeks), and (d) treatment effect ([−16, 16] weeks).

Table 6. The moderating effect of social media.

Column	(1)	(2)	(3)	(4)
Time window	[−16, 16] weeks	[−16, 16] weeks	[−16, 16] weeks	[−16, 16] weeks
Treated	−0.1366***	−0.2993***	−0.1008***	−0.6477***
	(0.0078)	(0.0119)	(0.0113)	(0.0176)
*Travala-post*Treated*	0.0450***			
	(0.0025)			
*Travala-include*Treated*		0.0546***		
		(0.0022)		
*Travala-post-like*Treated*			0.0153***	
			(0.0017)	
*Travala-include-like*Treated*				0.0892***
				(0.0024)
State-website trends	0.0561***	0.0557***	0.0560***	0.0548***
	(0.0009)	(0.0009)	(0.0009)	(0.0009)
Hotel fixed effect	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes
Obs.	1,949,632	1,949,632	1,949,632	1,949,632
R^2	0.6436	0.6437	0.6436	0.6439

Note: Robust standard errors (clustered by hotel ID) in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. The main effect variables (i.e., Travala-post, Travala-include, Travala-post-like, and Travala-include-like) are absorbed by the time-fixed effect.

Table 7. The moderating effect of cryptocurrency price.

Column	(1)	(2)	(3)	(3)
Time window	[−16, 16] weeks	[−16, 16] weeks	[−16, 16] weeks	[−16, 16] weeks
Treated	−0.0535***	−0.0536***	−0.0537***	−0.0529***
	(0.0029)	(0.0029)	(0.0029)	(0.0029)
*Mega*Treated*	0.1136***			
	(0.0087)			
*Large*Treated*		0.1273***		
		(0.0087)		
*Broad*Treated*			0.1316***	
			(0.0086)	
*AVA*Treated*				0.0207***
				(0.0042)
State-website trends	0.0080***	0.0080***	0.0080***	0.0079***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Hotel fixed effect	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes
Obs.	1,949,632	1,949,632	1,949,632	1,949,632
R^2	0.6397	0.6397	0.6398	0.6397

Note: Robust standard errors (clustered by hotel ID) in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. The main effect variables (i.e., Mega, Large, Broad, and AVA) are absorbed by the time-fixed effect.

(see Figure 3). In order to ascertain the gap between the check-out date and the review date, we collect the back-end data for all the Expedia Group (i.e., Expedia, Hotels.com, Orbitz, and Travelocity) hotels in California, Florida, and Texas (the data collection task was conducted in June 2022, and this data port was closed from September 2022). In order to prevent the data from being influenced by COVID-19 and be consistent with the time windows used in this study, we extract the check-in date for reviews posted four months before and after the event via the back-end data, and then match them with our original dataset (which only contains the review data and stay days obtained from the front-end data for both Booking.com and Expedia Group) using the review text and reviewer name. Finally, we calculate the gap as "review date minus check-in date minus stay days" and report its summary statistics in Table 8.

We find that about 42% of reviews are posted one day after checkout and more than 90% are posted within 6 days after checkout. We thus use lags from 0 to 7 days to calculate the weekly number of customer reviews to get robust empirical results. That is, the check-in date related to each review is "review date minus gap (ranges from 0 to 7) minus stay days."

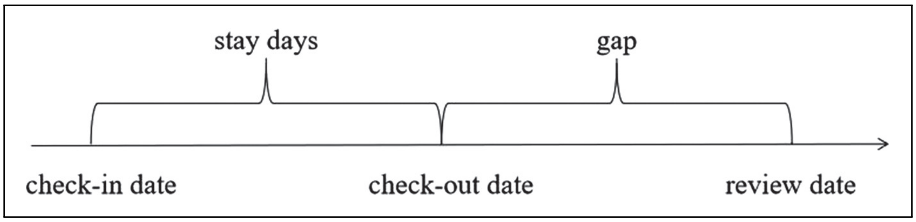

Figure 3. The generation process of consumer reviews.

Table 8. Summary statistics of the gap between review date and check-out date.

Lags	Frequency	Percentage	Accumulation
Lag = 0 day	30,711	11.67%	11.67%
Lag = 1 day	110,459	41.96%	53.63%
Lag = 2 days	37,722	14.33%	67.96%
Lag = 3 days	13,896	5.28%	73.24%
Lag = 4 days	7,486	2.84%	76.08%
Lag = 5 days	28,510	10.83%	86.92%
Lag = 6 days	9,408	3.57%	90.49%
Lag = 7 days	4,407	1.67%	92.17%

Table 9 reports the SDID estimation results with lags from 0 to 7 days when calculating the weekly number of consumer reviews. We find that these results are highly consistent with those reported in Table 5 (lag = 1 day). Therefore, our results are robust even though the dependent variable is calculated using different lags.

5.2 The Validity of Using Hotel Reviews as a Proxy for Hotel Revenue

In this study, we use the number of hotel reviews as a proxy for hotel revenue. Measurement validity is a critical concern. In order to address this concern, we collect the real revenue data of hotels in Texas and construct a dataset containing both hotel revenue and hotel reviews. First, we focus on the largest 14 cities (which have populations exceeding 200,000) in Texas and collect the revenue data of hotels in these cities using Search Texas Tax (a paid website that offers the monthly revenue data of hotels in Texas: https://app.searchtexastax.com). We obtain the revenue data of 17,133 hotels. Second, we put the address of each hotel into Google Maps API to unify the address format and match the hotels listed on both Booking.com and the Expedia Group platforms. A total of 735 hotels have revenue data and review data for 2019.

The final dataset contains two sets of variables: the monthly revenue data (*Revpar*: revenue per available room) of each hotel and the monthly number of consumer reviews (*Revnum*) of each hotel. As we do not have the booking date linked to each consumer review, the number of lags when calculating the monthly number of reviews to measure the contemporaneous revenue is a critical issue. We use up to 30 lags (0–29 days) to calculate the monthly number of reviews for each hotel and then regress it with *Revpar*. The results are reported in Appendix A. We observe that the adjusted R^2 is ~82%, suggesting that using the monthly number of consumer reviews to measure the monthly revenue of a hotel is reasonable. We also find that the magnitude of the coefficient of *Revnum* decreases as the lag increases. The coefficient drops from 0.137 to 0.127 (lag = 7 days) to 0.117 (lag = 14 days) to 0.096 (lag = 21 days) to 0.082 (lag = 28 days). These results suggest that the variable *hotel reviews* capture *hotel revenue* in a timely manner. Our empirical findings are consistent with existing findings that the majority of bookings are made within seven days of check-in. For example, according to a study by Hotwire.com in 2019, 84% of Millennials have booked a trip one week or less out from their departure day[4]; based on 12 million reservations from 600 hotels across the world, a study revealed that 48% of bookings made in 2018 are within seven days of check-in.[5] Therefore, our results in Section 5.1 using up to 7 lags are reasonable.

5.3 Seasonal Effect and Asymmetric Cooperation

Our results could be affected by potential seasonal effects in that consumers may prefer specific OTAs at certain times of the year (e.g., the festive season in December). In order to account for this, we conducted a placebo test using the same time windows (i.e., 4 weeks before and after 25th November) in 2018. The SDID estimation results are reported in column 1 of Table 10. We observe that the negative treatment effect seen in the results in Table 5 disappears. The positive coefficient (of a small magnitude) indicates that potential seasonal effects should not amplify the negative treatment effect induced by the event. Another concern is that the negative treatment effect may be caused by cooperation with small brands and not the adoption of a cryptocurrency payment. Although Travala.com is a leading cryptocurrency payment OTA, it is a small brand in the market of OTAs. Whether cooperation between large/well-known enterprises and small/growing enterprises benefits or harms the strong brand remains an open question (Besharat and Langan, 2014). We thus set an SDID design using the event that Booking Holdings cooperated with Getaroom[6] (a small brand in the market of OTAs) to expand market share on 12 November 2021.[7] The nature of the event is similar to that involving Travala; the only difference is that Getaroom does not use cryptocurrency payment. The estimation results in column 2 of Table 10 show a positive treatment effect.

Table 9. Synthetic difference-in-differences (SDID) estimation results using different lags.

Time window	[−4, 4] weeks	[−8, 8] weeks	[−12, 12] weeks	[−16, 16] weeks
Lag = 0 day	−0.1535***	−0.0899***	−0.0530***	−0.0387***
	(0.0044)	(0.0037)	(0.0035)	(0.0034)
Lag = 1 day	*−0.1393****	*−0.0814****	*−0.0466****	*−0.0354****
	(0.0044)	*(0.0037)*	*(0.0035)*	*(0.0034)*
Lag = 2 days	−0.1236***	−0.0653***	−0.0420***	−0.0310***
	(0.0044)	(0.0037)	(0.0035)	(0.0034)
Lag = 3 days	−0.1110***	−0.0533***	−0.0352***	−0.0240***
	(0.0044)	(0.0037)	(0.0035)	(0.0034)
Lag = 4 days	−0.0969***	−0.0419***	−0.0228***	−0.0191***
	(0.0044)	(0.0037)	(0.0034)	(0.0033)
Lag = 5 days	−0.0936***	−0.0368***	−0.0154***	−0.0129***
	(0.0044)	(0.0036)	(0.0034)	(0.0033)
Lag = 6 days	−0.0841***	−0.0278***	−0.0115***	−0.0096***
	(0.0044)	(0.0036)	(0.0034)	(0.0033)
Lag = 7 days	−0.0825***	−0.0190***	−0.0030	−0.0002
	(0.0044)	(0.0036)	(0.0034)	(0.0032)

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 10. Placebo tests of seasonal effect and cooperation with the small brand.

Column	(1)	(2)
Event	Seasonal effect	Small brand
Time window	[−16, 16] weeks	[−16, 16] weeks
Coef.	0.0152**	0.0467***
SE	0.0066	0.0036
t	2.303	12.97
Hotel fixed effect	Yes	Yes
Time fixed effect	Yes	Yes

Note: Robust standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

This suggests that our results in Table 5 are not induced by cooperation with a small brand/enterprise.

6 Experimental Study

While we recognize the technical significance of SDID in testing the treatment effect, it is important to acknowledge the limitations of relying solely on observational data. The endogeneity issue and the potential consumer flow from the treatment group to the control group are also important considerations. In order to strengthen the reliability of our findings, we incorporate experimental studies based on the work of Qiao and Rui (2023). Experimental studies offer several advantages. First, they allow for the direct manipulation of treatment variables, such as cryptocurrency payment adoption and social media externality, enabling us to establish a clearer causal inference (Wells et al., 2011). Second, the random assignment of participants in experimental studies significantly mitigates the influence of confounding factors, enabling us to attribute any observed disparities in firm performance with greater confidence (Shao et al., 2023). Third, we propose that the negative treatment effect is mainly driven by consumers' negative associations with cryptocurrency. Experimental studies are essential in understanding the underlying mechanisms of this phenomenon, as these studies allow us to observe the reactions of individual consumers (Serrano and Karahanna, 2016). Therefore, we conduct randomized experiments to check the validity of our findings derived from secondary data.

6.1 Research Design and Procedure

We use a between-subjects factorial design. The two factors investigated are cryptocurrency payment adoption (present vs. absent) and social media externality (positive vs. negative). Inspired by the growing body of experimental research, we utilize MTurk (Amazon Mechanical Turk) as a platform for conducting randomized experiments.[8] We randomly split 240 participants into one of the four treatment conditions (60 per group).[9] The participation is voluntary, and each participant receives 1$ for participating in the experiment tasks.

The task is designed using a vignette approach (Parboteeah et al., 2009), which effectively immerses participants in a scenario where they envision themselves as consumers of an OTA. Subsequently, participants are randomly assigned to one of four concise scenarios (a comprehensive description of the scenarios is provided in Appendix C). Upon completing the scenarios, each participant is asked to fill out a survey and will receive monetary compensation for their participation. We successfully recruit a total of 195 participants (for a response rate of 81.25%). The samples are well-balanced across all scenarios.[10]

We conduct a comparative analysis using multiple scenarios to examine how the two key factors—cryptocurrency payment adoption and social media externality—affect consumers' decision-making process. In order for us to establish

Table 11. Results of analysis of variance (ANOVA).

Constructs	Sum of squares	DF	Mean square	*F* value	Sig.
Cryptocurrency payment adoption	5.500	1	5.500	13.19	0.000
Social media externality	2.386	1	2.386	5.51	0.020
Cryptocurrency payment adoption × Social media externality	20.938	3	6.979	20.50	0.000

Note: DF represents the degree of freedom.

a baseline understanding of consumers' perceptions, the first scenario serves as the control group, presenting only a brief description of an OTA and its cooperation with a small OTA. In the subsequent scenarios, we maintain the same description as for the control group but add information about a collaboration with a leading cryptocurrency payment OTA (treatment group 1) and positive (treatment group 2) or negative (treatment group 3) feedback on social media. Finally, participants are required to respond to measures capturing their perceptions, manipulation checks, and demographics. In order to ensure the validity of the experiment procedures, instruments, and manipulations, a pretest involving postgraduate students ($n = 20$) and a pilot test of MTurk participants ($n = 50$) are conducted.

6.2 Measures

We employ consumers' continuance intention regarding a platform, which determines the performance and success of a platform (Bhattacherjee, 2001), as a proxy for the treatment effect. Continuance intention, based on Bhattacherjee (2001), is evaluated using a single item asking participants if they will continue using the OTA for hotel bookings after the cooperation event. Participants indicate their responses on a scale ranging from 1 ("*significantly reduce*") to 5 ("*significantly increase*"). Additionally, we ask participants two questions about the reasons why they decide to continue or not continue using the OTA. This approach aims to gain a deep understanding of consumers' attitudes to the OTA under different scenario treatments.

We employ different verification approaches to verify the power of manipulation. First, we assess the manipulation of cryptocurrency payment adoption by asking participants whether the business story mentions cryptocurrency payment technology and to what extent they agree or disagree with the statement that the OTA is related to cryptocurrency payment (on a 5-point scale). Second, following Liang et al. (2023), we include three questions to examine the validity of the manipulation of social media externality. Participants are asked to indicate whether the business story mentions the promotion of the cooperation through social media channels such as Twitter and the extent to which they agree or disagree with the statement that the cooperation event is promoted through social media channels. This ensures that participants' perceptions of the promotion of this event through social media are accurately captured. Additionally, participants are asked to respond to the question "To what extent do you agree or disagree with the statement that *cryptocurrency payment is a technology with more disadvantages than advantages*?" on a 5-point scale ranging from "*strongly disagree*" to "*strongly agree*." This question aims to assess participants' sentiments regarding cryptocurrency payment after they have been exposed to different social media scenarios, distinguishing between positive and negative sentiments (Chatterjee, 2019). The mean differences across treatments (i.e., four corresponding scenarios) are statistically significant (see Appendix B), confirming that the manipulations are successful.

6.3 Results

We conduct an ANOVA test to examine the effects of cryptocurrency payment adoption and social media externality on the continuance intention of consumers. We first perform Levene's test to assess the normality and homogeneity of the variance of the samples (Joseph et al., 2010). The p-values for these statistics are insignificant ($p > 0.05$), indicating that the distribution is normal and the variance is homogeneous. As shown in Table 11, the results demonstrate a significant ($F = 13.19$, $p < 0.001$) direct effect of cryptocurrency payment adoption on consumers' continuance intention regarding the target OTA. Furthermore, Table 11 reveals a significant ($F = 20.50$, $p < 0.001$) interaction between cryptocurrency payment adoption and social media externality, indicating that social media externality exerts a moderating effect on the relationship between cryptocurrency payment adoption and continuance intention.

The contrasting results in Table 12 present the detailed differences between the treatment groups and the control group in terms of continuance intention. Treatment group 1 has a significantly lower level of continuance intention than the control group, confirming the negative effect of cryptocurrency payment adoption on continuance intention. Treatment Group 2 (i.e., the adoption of cryptocurrency payment with positive social media promotion) shows a significantly higher level of continuance intention than Treatment Group 1, indicating that positive social media promotion can compensate for the negative impact of cryptocurrency payment adoption on continuance intention. Treatment Group 3 (i.e., the adoption of cryptocurrency payment with negative social media promotion) shows the lowest level of continuance intention of all the groups, suggesting that there is a significant decrease in continuance intention when consumers are exposed to negative information about cryptocurrency payment technology via

Table 12. ANOVA contrast results.

Contrast groups	Pairwise comparisons	SE	*t* value	Sig.	[95% conf. interval]	
Treatment 1 versus control	2.745 versus 3.104 (−0.359)	0.117	−3.06	0.013	−0.663	−0.055
Treatment 2 versus Treatment 1	3.106 versus 2.745 (0.361)	0.118	3.06	0.013	0.055	0.667
Treatment 3 versus Treatment 1	2.306 versus 2.745 (−0.439)	0.117	−3.76	0.001	−0.742	−0.136
Treatment 3 versus Treatment 2	2.306 versus 3.106 (−0.800)	0.119	−6.72	0.000	−1.109	−0.491

Note: Treatment 1 represents cooperation with a leading cryptocurrency payment OTA; Treatment 2 represents cooperation with a leading cryptocurrency payment OTA and positive feedback on social media; Treatment 3 represents cooperation with a leading cryptocurrency payment OTA and negative feedback on social media; the parentheses represent the mean difference; adjust *p*-values and confidence intervals for multiple comparisons using Tukey's method. OTA = online travel agencies; ANOVA = analysis of variance.

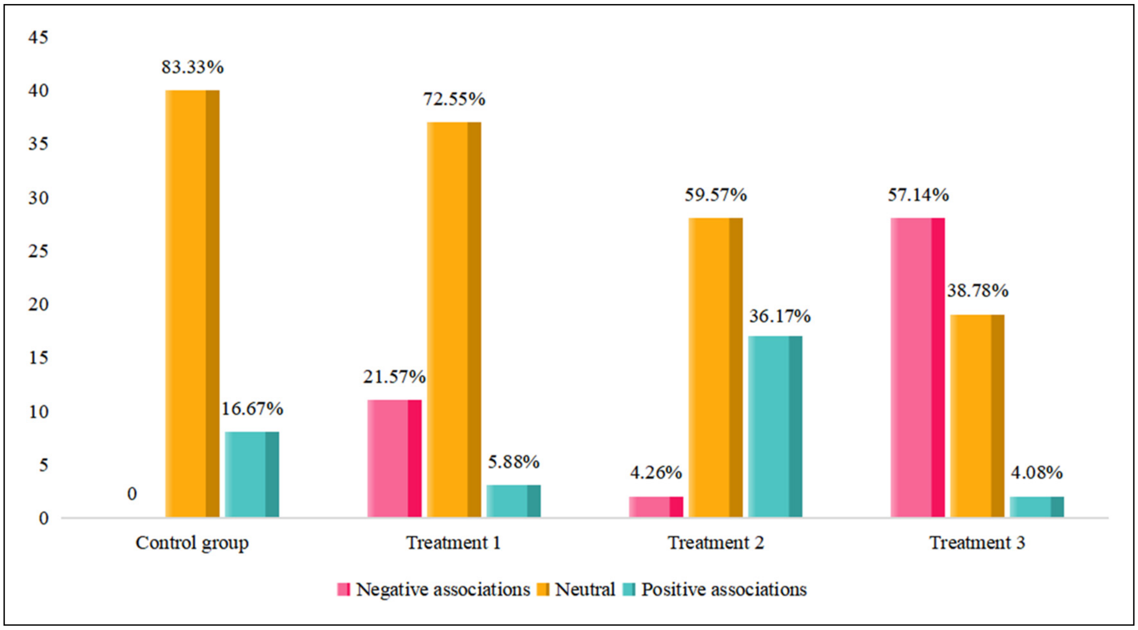

Figure 4. Coding results of the semi-structured questions.

social media. This finding provides additional support for the moderating role of social media externality.

We further code the responses to the semi-structured questions (see Footnote 1 of Appendix C) to uncover the underlying reasons for consumers' continuance intention. The initial question, featuring predefined choices, serves to minimize the likelihood of respondent misinterpretation, thereby reducing ambiguity and ensuring alignment between responses and intended meanings. The second question encourages respondents to furnish detailed and nuanced explanations for their choices, serving as a valuable tool to triangulate the findings obtained from the first question. As illustrated in Figure 4, the proportion of positive associations with the target platform regarding the cooperation event in treatment group 2 is the highest (36.17%) among the four groups. One of the respondents mentions: "Based on social media feedback, Twitter statements express positive associations with Company T's service, highlighting its novelty, fast transactions, and minimal fees. The cooperation event will positively influence my positive sentiment toward Company A, leading me to increase my bookings with the company." However, this phenomenon is reversed in treatment group 3, with participants expressing overwhelmingly (57.14%, the highest among the four groups) negative associations with the target platform regarding the cooperation event. These results indicate that negative associations caused by cooperation with a cryptocurrency payment platform reduce consumers' continuance intention regarding the target platform, as echoed by one respondent: "Regarding cryptocurrencies like Bitcoin, Ethereum, and others, I have significant skepticism, particularly when legitimate companies engage in NFTs or rely on third parties. Tweets correctly highlight the volatility and chaotic nature of an unstable market. This lack of faith leads me to hesitate in booking hotels with Company A, even though they do not directly accept crypto, but only through Company T."

Overall, the results of the experimental studies support our findings derived from observational data that the cooperation event induces negative associations with the target platform and hence decreases its online sales. However, social media functions as a powerful moderator in that positive promotion of the event via social media can alleviate the negative treatment effect.

7 Discussion and Conclusion

7.1 Main Findings

Leveraging a unique research context in which Booking.com cooperated with Travala.com to allow consumers to book all the hotels listed on Booking.com via the cryptocurrency payment platform, this study uses DID approaches and randomized experiments to identify the impact of the adoption of cryptocurrency payment on hotel performance. We have three main findings. First, the adoption of cryptocurrency payment via Travala.com induces a significant decrease in hotel sales on Booking.com. Second, the social media popularity of cryptocurrency payment and the price of cryptocurrency moderate this negative effect. Third, the driving mechanism of the negative adoption effect is consumers' negative associations with cryptocurrency.

7.2 Research Contributions

This study is one of the first attempts to empirically examine the effect of adopting cryptocurrency payment, which is a technology with social characteristics, on firm performance. By extending the associative learning model (Shimp, 1991; Washburn et al., 2004) to the context of technology adoption, our research indicates that the adoption of cryptocurrency payment could invoke negative associations from consumers and thus have an unexpected negative impact on firm performance. Although social media has become an important platform for firms to connect with customers and promote their products and services, the impact of social media on customer perceptions of controversial technology adoption is not always clear. This study demonstrates that social media mitigates the negative effect of adopting cryptocurrency payment on firm performance. These findings have practical implications for enterprises that plan to adopt technologies with social characteristics. First, firms should be cautious when adopting such technologies, as consumers may develop negative associations that negatively affect firm performance. Furthermore, firms can leverage social media to educate and engage with customers about the benefits and risks of cryptocurrency payment adoption and to address any concerns or misconceptions that may arise. By taking a proactive and strategic approach to cryptocurrency payment adoption and leveraging the power of social media, firms can position themselves to effectively navigate the rapidly changing landscape of digital payment systems and build consumer trust and loyalty to improve their overall performance.

7.3 Limitations

Our research has several limitations. First, cryptocurrency payment is not directly adopted by Booking.com in our research context. Because the adoption takes place indirectly through cooperation with Travala, our exploration of the long-term effect of the adoption is restricted. Second, there is a potential spillover effect of sales from Booking.com to Expedia Group platforms, which could lead to an overestimation of the magnitude of the adoption effect; hence, we conducted randomized experiments. Although we test the robustness of our results, the exact effect size is still difficult to estimate. Future research may find appropriate scenarios to explore the long-term effect and exact effect size of cryptocurrency payment adoption. Third, although existing findings suggested that the majority of bookings are made within seven days of check-in, the exact proportion is unknown because the booking date of each reservation is unavailable.

7.4 Conclusions

Emerging social technology has revolutionized business models and reshaped consumer behaviors. However, in the early stages of cutting-edge technology adoption (e.g., cryptocurrency, live streaming, and crowdfunding), inflated expectations followed by early disappointment often occur (Sodhi et al., 2022). The nature of cutting-edge technologies often makes their adoption controversial among the general public. Therefore, their adoption needs to be assessed from both an operational and a social perspective. This research provides an example of estimating the effect of adopting a novel technology—cryptocurrency payment. We establish the negative effect of this adoption on company sales and attribute the mechanism to consumers' negative associations. Moreover, by demonstrating the positive externalities of social media, we suggest a potential way to mitigate the negative consequences of controversial cutting-edge technology adoption.

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Notes

1. https://coinmap.org/view/#/world
2. https://seekingalpha.com/symbol/BKNG/press-releases https://www.nasdaq.com/market-activity/stocks/bkng/press-releases
3. https://seekingalpha.com/symbol/EXPE/press-releases https://www.nasdaq.com/market-activity/stocks/expe/press-releases
4. https://www.travelagentcentral.com/running-your-business/stats-84-millennials-have-booked-a-last-minute-trip
5. https://www.revinate.com/blog/your-average-guest-wont-book-more-than-a-week-in-advance/
6. https://www.getaroom.com/
7. https://www.phocuswire.com/booking-holdings-to-acquire-getaroom-for-1-2-billion
8. MTurk is an online marketplace that allows researchers to recruit participants for various tasks and experiments with diverse advantages, such as access to a diverse participant pool, cost-effective data collection, random assignment for establishing causal relationships, and the opportunity to measure constructs from consumers' point of view directly.
9. We perform a statistical power analysis to determine the minimum sample size required to test our hypotheses. Assuming an anticipated large effect size of 0.350, a desired statistical power level of 0.95, and a confidence level of 0.80, the required minimum sample size to estimate our model is 36 (Faul et al., 2009). Thus, our sample size of 60 for each group demonstrates sufficient statistical power.
10. There is no significant difference in gender (Pearson's chi-square value = 6.2375, $p = 0.101$) or age ($F = 0.36$, $p = 0.782$) distribution across the treatment conditions.

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